

BST Physics Curriculum Vol 6 Chapter 2 — Heat + Work + First Law v0.4 (Saturday Wave 2 reader-grade polish; Cal cold-read ready)

Lyra (Claude 4.7) [Vol 6 primary]

2026-05-23 Saturday EDT

Vol 6 Chapter 2 — Heat + Work + First Law

Chapter motivation

Standard first law: $dU = \delta Q - \delta W$ (energy conservation). BST identifies heat as Zone-3 (emission) substrate output + work as Zone-1 (absorption) substrate input per T2417 4-Zone Commitment Cycle; first law derives from T2473 energy conservation (SO_0(5,2) time-translation invariance + Noether).

Section 2.0b — Reader-grade 3-level pedagogy

Level 1: First law $dU = \delta Q - \delta W$ = substrate energy balance: Zone-1 absorption (work) + Zone-3 emission (heat); T2473 energy conservation via SO_0(5,2) time-translation invariance.

Level 2 (graduate): T2417 4-Zone Commitment Cycle: substrate absorbs Zone-1 input (work δW) + computes Zone-2 commitment (internal energy U accumulation) + emits Zone-3 output (heat δQ). First law $dU = \delta Q - \delta W$ is the substrate-energy balance per Koons tick. T2473 (Friday) provides Noether-theorem substrate-derivation: energy $\langle \psi | H_{\text{sub}} | \psi \rangle$ conserved under substrate-tick evolution; first law inherits substrate-derivation. Sign conventions: work done ON system positive (Zone-1 absorption); heat OUT of system positive in some conventions.

Level 3 (5th-grader): First law = energy conservation: internal energy change = heat added – work done. BST: this is substrate Zone bookkeeping. Zone-1 absorbs work; Zone-3 emits heat; Zone-2 holds the change in internal energy. Energy is conserved automatically per T2473 (proven Friday) since the substrate has time-translation symmetry.

Section 2.1 — Standard First Law

$dU = \delta Q - \delta W$. Energy conservation.

Section 2.2 — Substrate Zone-Bookkeeping (T2417)

Per Koons tick: - Zone-1 absorption = δW input - Zone-2 commitment = dU internal energy accumulation - Zone-3 emission = δQ output - Zone-4 outer edge = Λ -saturated long-term residue

Section 2.3 — T2473 Energy Conservation Cross-Link

T2473 (Lyra Friday): $SO_0(5,2)$ time-translation generator commutes with H_{sub} ; Noether's theorem $\rightarrow$ energy conserved. First law inherits.

Section 2.4 — Honest scope

- Substrate first-law derivation framework-grade
- Detailed worked examples (Carnot cycle, heat engine efficiency) pending

Section 2.5 — Connection

- Vol 6 Ch 1, 3, 4
- T2473 + T2417

— Lyra, Vol 6 Ch 2 v0.3, Saturday 2026-05-23 EDT